

Principal Angles of Two Finite Nuisance Panels: A Grassmann Stability Audit

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Abstract

The preceding finite panel-contrast audit found a narrow raw-weighted residual crossing, but the crossing disappeared under equal-row weighting. We remove the nuisance-coordinate choice and compare the two panel nuisance column spaces by principal angles and orthogonal projectors. On the locked TPC-341/TPC-342 response panels, the raw principal cosines are 0.9957018010 and 0.0799456793, while equal-row weighting gives 0.9144519860 and 0.0787084493. Thus one direction is strongly aligned but a second direction is nearly orthogonal, and the leading angle changes from 5.3142 to 23.8720 degrees. A fixed nonsingular coordinate shear preserves the projectors and cosines to numerical precision, confirming that the separation is a subspace statement rather than a basis artifact. Mutual cross-panel target transfer fails a residual-retention threshold in both weightings, and all 18 leave-one-control-out angle pairs preserve the transverse separation. This is a finite geometric obstruction only; it provides no arithmetic, asymptotic, Route-A/Route-B, or twin-prime result.

1 Question and scope

TPC-344 added a signed panel-contrast column to each nuisance category and found a raw pooled residual retention just below 0.30. Its equal-row value was above 0.30, so the apparent repair could reflect row weighting rather than a stable relation between the two panels. The present paper asks the coordinate-invariant version of that question: *what is the geometry of the two nuisance subspaces themselves?*

All objects in this paper are finite vectors generated by the repository’s hash-locked source and operator implementations. The labels Twin and nuisance name declared masks and response vectors; they are not an arithmetic decomposition. In particular, a principal angle between two finite response subspaces is not a source-uniform estimate and does not pay any Route-B loss budget.

2 Frozen finite protocol

The two inherited panels are

$$\begin{aligned}\mathcal{P}_{341} &= [48097, 48608], [48609, 49120], [49217, 49728], \\ \mathcal{P}_{342} &= [40097, 40608], [40609, 41120], [41121, 41632].\end{aligned}$$

Every row has source length 512 and scale 1024. The response operator is the all-plus deleted-diagonal operator with $Q = 54$, kernel exponent one, and height $H = 66$. Nine fixed coordinate bijections act on four source masks: twin-prime, non-twin-prime shift, prime-power shift, and zero support.

For a panel p and nuisance category j , let $n_{p,r,j}$ be the nine-control mean output in row r , and let $y_{p,r}$ be the corresponding twin-prime mean. The nuisance matrix is formed by vertically stacking the three $n_{p,r,j}$ columns:

$$N_p = \begin{bmatrix} n_{p,1,1} & n_{p,1,2} & n_{p,1,3} \\ n_{p,2,1} & n_{p,2,2} & n_{p,2,3} \\ n_{p,3,1} & n_{p,3,2} & n_{p,3,3} \end{bmatrix},$$

where each displayed entry is a length-512 block. The target Y_p is the analogous stack of the three $y_{p,r}$ vectors. For the declared equal-row sensitivity audit, every block in row r is divided by $\|y_{p,r}\|_2$ before stacking. Raw weighting uses no such rescaling.

3 Finite Grassmann geometry

Let

$$N_p = Q_p \Sigma_p V_p^T$$

be a positive-SVD factorization, with the columns of Q_p retained according to the declared floating-point rank rule. Write $P_p = Q_p Q_p^T$. The two nonzero singular values of $Q_{341}^T Q_{342}$ are denoted $c_1 \geq c_2 \geq 0$ and their principal angles are $\theta_i = \arccos(c_i)$. The observed ranks are three and two, so two angles are reported and one direction is panel-specific.

Proposition 1 (principal-angle invariance). *The singular values of $Q_1^T Q_2$ depend only on the two finite column spaces, not on the choice of orthonormal bases or on nonsingular reparameterizations of the original nuisance columns.*

Proof. If $\tilde{Q}_i = Q_i U_i$ for orthogonal matrices U_i , then

$$\tilde{Q}_1^T \tilde{Q}_2 = U_1^T (Q_1^T Q_2) U_2,$$

which has the same singular values. If N_i is replaced by $N_i S_i$ for nonsingular S_i , its column space and hence its orthogonal projector are unchanged. Choosing any orthonormal bases of those unchanged spaces gives the same principal cosines. $\square$

Proposition 2 (finite projection identity). *For every finite target Y and orthogonal projector P ,*

$$\|Y\|_2^2 = \|PY\|_2^2 + \|(I - P)Y\|_2^2.$$

Proof. The two vectors PY and $(I - P)Y$ are orthogonal and sum to Y . The identity is therefore the Pythagorean theorem. The implementation evaluates the two energies and checks the numerical decomposition gap. $\square$

For a target from one panel projected onto the other panel's nuisance space, we record the finite residual retention

$$\rho(Y, N) = \frac{\|(I - P_N)Y\|_2^2}{\|Y\|_2^2}. \quad (1)$$

The mutual-transfer criterion used here requires both directions to have $\rho < 0.30$. This is a declared finite diagnostic, not a theorem about future source intervals.

Table 1: Main finite Grassmann geometry and cross-panel transfer.

quantity	raw	equal-row
TPC-341 nuisance rank	3	3
TPC-342 nuisance rank	2	2
principal cosine c_1	0.9957018010	0.9144519860
principal cosine c_2	0.0799456793	0.0787084493
principal angle θ_1 (degrees)	5.3141837613	23.8719787000
principal angle θ_2 (degrees)	85.4145566103	85.4856687739
$\ P_{341} - P_{342}\ _F$	1.7333127887	1.8207594818
$\rho(Y_{341}, N_{342})$	0.2306119635	0.2745950088
$\rho(Y_{342}, N_{341})$	0.3588770900	0.3234520500

4 Certified readout

The producer generates $216 = 2 \cdot 3 \cdot 9 \cdot 4$ raw records, of which 171 are nonempty. The principal-angle and transfer results are shown in Table 1.

The raw leading cosine exceeds the predeclared 0.99 alignment threshold, and both second cosines are below 0.20. Thus the finite geometry has one near-common direction and one nearly transverse direction. The second direction is not a numerical zero: it corresponds to angles near 85.4 degrees, while the rank difference leaves one additional TPC-341-only direction.

The first angle changes by

$$23.8719787000 - 5.3141837613 = 18.5577949387 \text{ degrees.}$$

This exceeds the predeclared 10 degree sensitivity threshold. The principal-angle readout therefore gives an independent, basis-invariant reason not to promote the TPC-344 raw crossing to a weighting-stable law.

5 Coordinate and control robustness

To test coordinate dependence, both nuisance matrices are multiplied on the right by the fixed nonsingular shear

$$S = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

In the raw audit the maximum entrywise projector change is 7.65×10^{-15} and the maximum principal-cosine change is 5.56×10^{-16} . The equal-row errors are smaller than these values. The observed ranks remain (3, 2). This is a numerical check of the exact coordinate-invariance proposition, not a claim that the finite spaces are canonical arithmetic objects.

As a hostile control check, each of the nine controls is omitted in turn. The nuisance means are formed from the other eight controls and the omitted twin output is used as the target block. The raw leading-cosine minimum over the nine pairs is 0.9947001951, and the raw transverse-cosine maximum is 0.1549751276. Under equal-row weighting the transverse maximum is 0.1675760096. Thus the near-orthogonal transverse direction persists over all 18 leave-one-control-out geometry pairs.

The transfer readout is deliberately asymmetric. TPC-341 targets projected onto the TPC-342 space have residual retentions 0.2306 and 0.2746 under the two weightings, whereas TPC-342 targets projected onto TPC-341 have retentions 0.3589 and 0.3235. Since the mutual criterion requires both directions

below 0.30, it is refuted in both finite weightings. These numbers are orthogonal projection residuals and do satisfy Proposition 2; they are distinct from coefficient cross-fit prediction errors.

6 Independent certification and claim firewall

The producer locks the TPC-340 source/operator parent, the TPC-341 and TPC-342 panel certificates, and the TPC-344 route parent. A reverse-shell checker does not import the producer. It uses a separately hash-locked reverse engine, reconstructs all masks and controls, and compares every raw-record label and energy plus the principal-angle, projector, transfer, and leave-one-control-out metrics. A nine-mutation stress suite rejects changes to the protocol, geometry thresholds, transfer obstruction, invariance flag, rank census, arithmetic firewall, and exact anchor.

Table 2: Claim ledger.

item	status
principal-angle definition and invariance	proved exact finite model
projection/Pythagorean identity	proved exact finite model
raw dominant alignment and transverse separation	numerically certified finite scoped
leave-one-control-out transverse separation	numerically certified finite scoped
weighting-stable common nuisance law	refuted scoped
mutual cross-panel target transfer	refuted scoped
source-uniform arithmetic L^2	open
uniform masked operator bound	open
full Route-B Gate B	open
fixed-power credit	0
twin-prime conclusion	none

The Session-named Route-A and Route-B evaluator files are absent in this checkout. The local Bridge-B wrapper is consequently a fail-closed fallback and cannot be called an official evaluator pass. No finite principal-angle certificate supplies an arithmetic saving or the strict 1/400 endpoint.

7 Conclusion and next question

The TPC-344 panel-adaptive repair has a clear finite geometric profile: one dominant direction is almost shared under raw weighting, a second direction is nearly orthogonal in both weightings, and the dominant angle is strongly weight-sensitive. The coordinate-shear test shows that this profile is not caused by the selected nuisance coordinates. The asymmetric transfer failure prevents a mutual common-law interpretation, while the leave-one-control-out audit preserves the transverse obstruction.

The reusable route is

$$\begin{aligned} &\text{panel-adaptive span} \longrightarrow \text{coordinate-invariant subspaces} \\ &\longrightarrow \text{principal angles} \longrightarrow \text{weighting/transfer obstruction.} \end{aligned}$$

The next minimal question is a finite no-go or freeze test: determine whether adding further panel-specific degrees of freedom produces a robust model class or merely overfits these locked panels. Until a source-uniform arithmetic theorem is supplied, Route-A, Route-B, and the twin-prime endpoint remain open.